OPSEARCH (2018) 55:478–501
https://doi.org/10.1007/s12597-018-0331-3

APPLICATION ARTICLE

Prioritization of project proposals in portfolio management using fuzzy AHP

Kajal Chatterjee[1] · Sheikh Ahmed Hossain[2] · Samarjit Kar[1]

Accepted: 14 January 2018 / Published online: 22 January 2018
© Operational Research Society of India 2018

Abstract Strategies in project assortment and prioritization directly influence organizational production and cost-effectiveness. Due to dearth of funding and nominal technology with improper judgments of expert's, selection of optimal project portfolio can be viewed as a risk based decision making problem considering various risk factors. Recently, fuzzy set theory is applied for modeling real world problems due to failure of traditional models in uncertain environments. This paper deals with project proposal prioritization from a set of project portfolio satisfying a set of criteria evaluated by decision makers. The analytic hierarchy process is applied in fuzzy environment to help the management of any project based company to prioritize in terms of investment. In the decision approach, expert team decides whether to accept or reject a project as per set of criteria and sub-criteria based on diverse project risk levels. A hierarchy among the different projects based on ratings clearly prioritize the projects among the suggested proposals.

Keywords Project selection · Fuzzy set · Portfolio management · Analytic hierarchy process · Multi-criteria decision making (MCDM)

✉ Kajal Chatterjee
chatterjeekajal7@gmail.com

Sheikh Ahmed Hossain
skahossainmath@gmail.com

Samarjit Kar
kar_s_k@yahoo.com

[1] Department of Mathematics, National Institute of Technology, Durgapur 713209, India

[2] Department of Mathematics, Brahmananda Keshab Chandra College, Kolkata 700108, India

1 Introduction

In a project based organization, proper selection of projects and resource allocation are critical issues where complexity in decision process arises when the number of projects increases. From among the available project proposals, organizations with resource constraints need to prioritize projects when implemented [1]. As per Tavana et al. [2], project portfolio selection is a process of selecting a portfolio of projects from available projects proposals without surpassing available organizations assets or violating organizational limitations. Several reputed firms are getting several projects as most of the agencies intend to outsource their projects to them. Turner [3] suggested that organization must select project portfolio that are beneficial, *in priority basis*, due to insufficient resources or expertise to fund their projects and make optimal use of their available human and material resources to grab every opportunities. Also, companies must concentrate on their market reputation as decision taken emotionally in selection of projects, without evaluating ifs and buts of project, may cause serious damage to its image in market [4]. Thus companies should undergo through research on selection of projects analyzing both technical and fundamental parts, thereby ensuring selection of most feasible projects.

Project supervisors usually adopt priority system-guidelines for several alternatives to balance the competing demands. Major decisions are generally affected and more effective when made in timely and proficient manner [5]. Proper choices are undertaken in exercising best choice from among multiple project portfolios, taking both physical and financial constraints of the firm into consideration [6]. The organizational decision makers develop guidelines for selecting models that minimize financial risks while maximizing the likelihood of success [7]. Although, a number of decision models are available to managers for evaluating potential projects, business firms opt to develop a screening model to prioritize projects taking financial constraint into account. As per Shouder [8], managers should consider five important issues, namely, *realism, flexibility, capability, ease of use and cost*, when evaluating screening models. The main target of project selection is scrutinizing project sustainability for approval or rejection, following a set of regulated stages and checkpoints [9]. Some of the objectives of the project portfolio management are *identification, ranking, prioritization, selection and authorization of projects* [10]. Uncertainty and volatility are increasing day by day and managers take strategic decisions on project portfolio (*like tender participation*) under non-deterministic conditions [11]. The complication in project selection problem arise while choosing a subset (*portfolio*) from among a number of projects [12]. In real world multicriteria decision making (MCDM) problems, a decision maker (DM) often selects or prioritize alternatives linked with intangible or incompatible attributes. Companies should align project portfolio with their strategic business objectives, combining performances of its components in order to maximize the shareholders' value while balancing resources allocation and risks. Thus, project selection can be defined as a complex decision making process, influenced by multiple conflicting objectives.

As project success is the ultimate objective of a company, critical success factors (CSFs) affecting its future implementation should be the pillars of the selection criteria [13]. Gutjahr et al. [14] developed a tool for project evaluation based on proficiency, extending it to multi-criteria decision analysis due to its mathematical complication. Chu et al. [13] applied a heuristic method based on a fuzzy logic for grading project portfolio. The problem for optimal project funding implies decisions on the new projects and on the projects to be continued. Shpak and Zaporojan [15] and Danila [16] briefly undergone literature review of existing project selection methodologies and discussed various multi-criteria aspects of the problem. On non-financial criteria, Halouani et al. [17] developed PROMETHEE-MD-2T MCDM method but was difficult in developing a selection model for small and medium enterprises. As per different industries and environments, project managers have to identify the most opportune set of critical success factors, trying to implement the right practices that satisfy all the stakeholder's requirements. Tiryaki and Ahlatcioglu [18] examined project selection methodologies and role of 'critical success factors' and 'key performance indicators (KPI)' in the project selection process, thus deepening the project implementation profile model. In project portfolio management, probable projects prioritization and selection encompass optimization algorithms or management techniques using explicit project selection criteria [5].

Evaluating optimal project portfolio value for the resources is often difficult due to multiple selection criteria, subjective and imprecise assessments and projects interdependencies [10]. Various proposals are undertaken to select optimum portfolios of projects. Wang and Hwang [19] proposed a methodology based on optimized value and strategic balance in selecting fuzzy portfolios of research and development (R&D) projects. Coffin and Taylor [20] presented a MCDM methodology using fuzzy logic and standard beam search for selecting and scheduling R&D investments. Majority of researchers has given more weightage to project uncertainty while selecting portfolios of projects [21]. Huang et al. [22] applied fuzzy analytic hierarchy process (FAHP) for R&D portfolio selection considering project uncertainty. Tiryaki and Ahlatcioglu [18] opted for fuzzy-AHP in choosing stock portfolios in the stock exchange market. Chen and Cheng [21] proposed a fuzzy MCDM methodology in ranking portfolios of information sourcing (IS) projects under uncertainty conditions. In this condition, we are applying fuzzy AHP for project prioritization in portfolio management taking into consideration certain measures discussed in later section.

The rest of the paper is structured as follows. Section 2 briefly summarize the framework and project selection methodologies in project portfolio management. Section 3 discuss the back ground of fuzzy AHP and extent analysis algorithmic methodology. Section 4 evaluate the hierarchical structure for prioritization of project portfolio, discussing proposed criteria and sub-criteria along with project alternatives. Computational procedure using the proposed approach is given in Sect. 5 along with result discussion. Finally, Sect. 6 draws some general conclusions along with future scope.

2 Project portfolio selection

2.1 Background framework

For the past decades, project management research is trying to improve the ability of organizations for implementing success in projects [23]. Project portfolio management aligns 'successful projects' with strategic business objectives, but 'expected project success' remains the main determinant for project selection, if shareholders' value is maximized while balancing resource allocation [11]. Therefore, project selection is a process of strategic significance aimed to assess single projects or groups of projects implementing a set of them to accomplish the purposes of the parent organization [1]. However, too often it fails due to complexity in project portfolio management caused by factors like indecision, interrelationships among projects and critical factors that are difficult to measure [10].

Managers can properly control project's portfolio and balance overall exposure to risks, with a clear perception of expected results on every single project [11]. For different kinds of industries and environments, project managers have to identify the most opportune set of *criteria success factors* (CSFs) and implement the right practices to satisfy stakeholders' requirements. The process of project assessment for project portfolio selection should always consider criteria, sub-criteria and key performance indicators. Factors are independent variables of a project that organizations can drive, while key performance indicators are significant dependent variables that measure outcomes and performances of the project [24]. Criteria are the basis to express judgments, whereas sub-criteria are the influential forces that contribute to the success or failure of the project. Considering the result of KPI, success or failure of a project is determine through the lens of criteria [25].

2.2 Project selection methodologies

For eliminating candidate projects, averaging below threshold values in respect to preset indicators, portfolio management use scoring models like multi-criteria analysis from varied tools and techniques [26]. In project portfolio management, prioritization and selection of possible projects, usually involve optimization algorithms or management techniques using specific selection criteria [5]. Mohanty [27] offered a multi-criteria decision-making technique in the selection of project proposals. Tumer and Muller [28] revealed the complexity of identifying the set of factors through an extensive survey, and showed variation in success or failure with respect to diverse features like selection of criteria, organizational structure, size of project, industrial sector, different perspectives of the stakeholders, and different stages of the life cycle.

A project portfolio is an assortment of projects put together for a particular organization. These projects generally compete for limited resources (e.g., *individuals, capitals and time*). Salo et al. [29] review extensively the literature on portfolio decision analysis, which provides a sound methodological basis to account for the complex environment faced by decision makers (DMs) while

allowing for the best possible resource allocation decision. Carlsson et al. [30] presented a fuzzy mixed integer programming model for the R&D optimal portfolio selection problem. Ho et al. [31] suggested a hybrid DEMATEL–ANP–VIKOR methodology in portfolio selection based on CAPM. Jeng and Huang [32] projected a hybrid MCDM method based on modified Delphi, decision-making trial and evaluation laboratory (DEMATEL), and analytic network process (ANP) for the project portfolio-selection problem. Khalili-Damghani and Sadi-Nezhad [33] proposed a framework using a goal programming (GP) and fuzzy TOPSIS using hybrid fuzzy multiple criteria group decision making approach for sustainable project selection. Liesio and Punkka [34] provide baseline value specification and sensitivity analysis in multi-attribute project portfolio selection. Abbasi et al. [35] selected balanced portfolios of R&D projects with interdependencies using a cross-entropy based methodology. An et al. [36] developed a multi-criteria decision making method by integrating interval AHP and interval VIKOR method for China's stakeholders to select the most efficacious portfolio for solving the severe problems caused by the informal e-waste recycling and promote the development of China's WEEE recycling industry in a sustainable approach. Smith-Perera et al. [37] proposed a project strategic index proposal (PSI) for portfolio selection in electrical company based on the analytic network process. Project portfolio selection (PPS) methods involve simultaneous consideration and ranking of a numbers of projects according to particular criteria. The most highly ranked projects are then selected for a portfolio without exceeding available organizational resources.

3 Fuzzy analytic hierarchy process (FAHP)

3.1 Background and review of fuzzy AHP

Analytic hierarchy process (AHP), a widely known MCDM tool is effectively applied to many practical real-life problems [22]. But this method cannot handle the inherent uncertainty and vagueness in mapping decision maker's (DMs) opinions to crisp numbers. Traditional AHP requires crisp judgments, but decision makers prefer fuzzy ones owing to intricacy and imprecision involved in real world decision problems [30]. For similar words, individual judgments of events are perpetually subjective with different clarifications. For well-defined word (*linguistic comparison of AHP questionnaire response*), the boundary criterion that determines the belongingness to the set, defined by that word is often fuzzy or vague. So, fuzzy numbers and fuzzy sets characterize linguistic variables [19]. The preferences in AHP are human judgments for intangible criteria (*based on social opinions*), allowing fuzzy approach for a more accurate description of the decision-making process [42]. Fuzzy AHP used to express subjective judgments of experts by pair-wise comparison matrix with earliest work presented by Van Laarhoven and Pedrycs [44] comparing fuzzy ratios with triangular membership functions. Various studies for fuzzy AHP [43–49] are suggested for decision making problems.

3.2 Extent analysis method (EAM) for priority weight

Since the publication of Saaty's [7] 'The analytic hierarchy processes', academics are working on its extension in fuzzy domain. Van Laarhoven and Pedrycs [44] suggested a method for expert's judgment using fuzzy numbers with triangular membership function (TFN) and simple operation laws. In this paper we prefer 'Extent analysis method', proposed by Chang [38] to handle fuzzy AHP, different from methods [43–49] with same ordering of permutation of elements. Firstly, the TFNs are used for pair-wise comparison scale of AHP. Then using, extent analysis method [50] in fuzzy domain, the *synthetic extent value of the pair wise comparison* are developed. Lastly, applying principle of comparison of fuzzy numbers [38], weight vectors for each element (sub-criteria) under a definite criterion is found out. The detail algorithmic methodology is given below.

3.2.1 First step

Let $X = \{x_1,\ x_2, \ldots, x_n\}$ be an object set, and $U = \{g_1,\ g_2, \ldots, g_m\}$ be a goal set. As per extent analysis method, each object is taken and extent analysis for each goal g_i is performed, respectively. So, M extent analysis values for each object are obtained, as:

$$M_{g^i}^1, M_{g^i}^2, M_{g^i}^3, \ldots, M_{g^i}^m; \quad i = 1, 2, \ldots, n$$

where all $M_{g^i}^j,\ \ (j = 1, 2, \ldots, m)$ are triangular fuzzy numbers.

3.2.2 Second step

The value of fuzzy synthetic extent with respect to the ith object is defined as:

$$S_i = \sum_{j=1}^{m} M_{g^i}^j \otimes \left[\sum_{i=1}^{n}\sum_{j=1}^{m} M_{g^i}^j\right]^{-1}; \quad i = 1, 2, \ldots, n$$

To obtain $\sum_{j=1}^{m} M_{g^i}^j$, perform the fuzzy addition operation of 'm' extent analysis values for a particular matrix such that:

$$\sum_{j=1}^{m} M_{g^i}^j = \left(\sum_{j=1}^{m} l_j,\ \sum_{j=1}^{m} m_j,\ \sum_{j=1}^{m} n_j\right)$$

And to obtain $\left[\sum_{i=1}^{n}\sum_{j=1}^{m} M_{g^i}^j\right]^{-1}$ perform the fuzzy addition operation of $M_{g^i}^j,\ \ (j = 1, 2, \ldots, m)$ values such that $\left[\sum_{i=1}^{n}\sum_{j=1}^{m} M_{g^i}^j\right] = \left(\sum_{j=1}^{n} l_j,\ \sum_{j=1}^{n} m_j,\ \sum_{j=1}^{n} n_j\right)$. And compute the inverse of $\left[\sum_{i=1}^{n}\sum_{j=1}^{m} M_{g^i}^j\right]$ such that: $\left[\sum_{i=1}^{n}\sum_{j=1}^{m} M_{g^i}^j\right]^{-1} = \left(\frac{1}{\sum_{j=1}^{n} l_j}, \frac{1}{\sum_{j=1}^{n} m_j}, \frac{1}{\sum_{j=1}^{n} n_j}\right)$

3.2.3 Third step

The degree of possibility of $M_2 = (l_2,\ m_2,\ u_2) \geq M_1 = (l_1,\ m_1,\ u_1)$ is defined as

$$V(M_2 \geq M_1) = Sup\left[\min\left(\mu_{M_1}(x), \mu_{M_2}(y)\right)\right]$$

which is equivalently expressed as follows:

$$V(M_2 \geq M_1) = hgt(M_2 \cap M_1) = \mu_{M_2}(d) = \begin{cases} 1, & if\ m_2 \geq m_1 \\ 0, & if\ l_1 \geq u_2 \\ \dfrac{(l_1 - u_2)}{(m_2 - u_2) - (m_1 - l_1)}, & otherwise \end{cases}$$

where d is the ordinate of the highest intersection point D between μ_{M_1} and μ_{M_2}.

To compare M_1 and M_2 we need both the values of $V(M_1 \geq M_2)$ and $V(M_2 \geq M_1)$. The degree of possibility for a convex fuzzy number to be greater than 'k' convex fuzzy numbers $M_i (i = 1, 2, \ldots, k)$ defined by:

$$V(M \geq M_1, M_2, \ldots, M_k) = V[(M \geq M_1)\ and\ (M \geq M_2)\ and \ldots (M \geq M_k)];\quad i = 1, 2, \ldots, k$$

Assume $d'(A_i) = \min V(S_i \geq S_k)\ for\ k = 1, 2, \ldots, n;\ k \neq i$. Then the weight vector W' is given by

$$W' = \left(d'(A_1),\ d'(A_2), d'(A_3) \ldots d'(A_n)\right)^T,$$

where $A_i (i = 1, 2, \ldots, n)$ are n elements. Via, normalization, the weight vectors W (*non-fuzzy number*) are evaluated as follows:

$$W = (d(A_1),\ d(A_2), d(A_3) \ldots d(A_n))^T$$

4 Evaluating hierarchical structure for prioritization project portfolio

In recent years, project selection problem is approached through fuzzy set theory [39], where linguistic terms [40] are expressed in triangular fuzzy numbers (TFNs) to model uncertainty. In this paper we apply Chang's *extent analysis method* [38] based FAHP approach aiding project prioritization in project portfolio management, as detailed in Sect. 3.2. The proposed approach evaluates alternative projects proposals involving subjective judgment involving three basic stages as follows: (1) Fix the goal, (2) Identify the criteria and sub-criteria to be used in the model, (3) Prioritization of alternative project proposals and determining the final rank.

4.1 Building the hierarchy model based on criteria and sub-criteria

For evaluating and prioritize project portfolio based on criteria and sub-criteria drawn out of related literature (*details given in* Sect. 4.3), primarily we establish a hierarchy model of project portfolio, as shown in Fig. 1.

We asked views of experts (*through e-mail and questionnaire survey mode*) from different areas of projects portfolio management to review the hierarchy model. Middle to senior level professionals of the case projects, having experience of *more than 10 years*, were preferred as experts for collection of their opinions. Thus, a team of 15 project management experts was formed consists of *two* senior

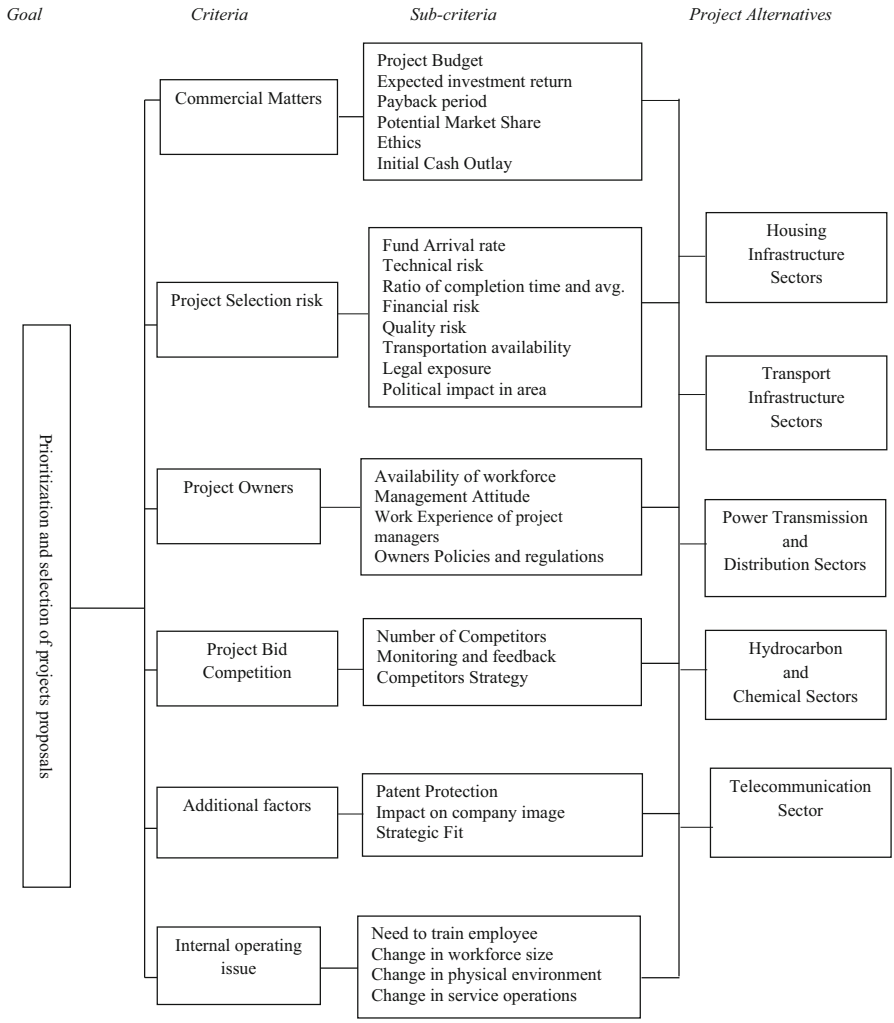

Fig. 1 The hierarchy model of investment project

managers, *three* project risks analyst and *two* purchasing manager and others specialized in project portfolio management, as detailed in Table 1.

All experts are capable to decision-making and having expertise in various managerial functions within the industry, i.e., in purchasing, planning, production, finance, quality, inspection, environmental management, etc. Before starting the work, the objectives of this study were explained to the experts. Besides, how the data will be used was also discussed with them. These experts were interviewed personally for collecting the necessary qualitative and quantitative data needed for this study. The majority in their decision is taken as common consensus among them and taken as granted unanimously.

Thus a hierarchy of four level evaluation index systems is presented as: Goal, Criteria, Sub-criteria and Alternatives, detailed in Fig. 1 for this paper. The goal of

Table 1 Summary of demographic profile about respondents

Characteristics	Frequency	Percentage (approx.)
Age group		
21–30	1	7
31–39	3	20
40–47	6	40
48–58	5	33
Gender		
Female	6	40
Male	9	60
Level of education		
Bachelor's	5	33
Master's	7	47
Higher	3	20
Role of respondents		
Chief personal officer	1	7
Manager or general manager	2	13
Staff or assistant manager	1	7
Project risks analyst	3	20
Purchasing manager (Finance section)	2	13
Construction site Engineer	3	20
Environmental representative	1	7
Quality inspection representative	2	13
Years of experience in construction sector		
Above 15 years	4	26
11–15 years	10	67
5–10 years	1	7
Less than 5 years	0	0
Total available number	15	

this paper is to prioritize suitable projects proposals depending on financial constraints and present resources of the company. The primary list of criteria and sub-criteria are updated and can be revised and altered as per organization culture and needs. Some sub-criteria may be added and some considered as unimportant can be eliminated, depending on present scenario. In a similar brainstorming methodology by the experts, suitable projects alternatives are also chosen.

4.2 Proposed criteria of project selection

Various researchers explored factors that have influence on project selection. For example, Mohanty et al. [12] pointed out four features namely, minimum investment, low complex technology, short period and high returns that a potential project should have. Lin and Chen [41] recommended that the contractor should consider the factors like essence bidding, value of tendering opportunities, resources and corporate reputation (*brand value*) in project selection. As per Gabriel et al. [51] project selection belongs to strategic level for any organization subjective to environmental and social factors. Wang et al. [4] suggested that during project selection, comprehensive evaluation are divided into two parts: the bid (non-bid) and in which project to bid. Pinto and Slevin [52] suggested ten key factors namely, project mission, top management support, project schedule, client consultation, personnel, technical tasks, client acceptance, monitoring and feedback, communication and troubleshooting in implementation of project portfolio management. Huang et al. [22] have taken four aspects, 11 objectives and 30 criteria. Meredith and Mantel [53] have taken five factors namely, production factors (14 *sub-factors*), marketing factors (8 *sub-factors*), financial factors (7 *sub-factors*), personnel factors (7 *sub-factors*), administrative and miscellaneous factors (8 *sub-factors*) respectively. Jiang and Klein [54] in their research into selection criteria for information systems (IS) projects generated six sub-categories of project evaluation criteria like financial, organizational, competing environment, technical, risk and management taking seven criteria. The range of criteria begins to align under supervision of standard of portfolio management [26]. In their study of strategic project selection for construction projects (*in ministry of Defense in Thailand*), Puthamont and Charoenngam [55], identified the wide range of project selection criteria for projects, namely, general project selection (10 *criteria*), is project selection (6 *criteria*), construction project selection (10 *criteria*), infrastructure project selection (9 *criteria*), R&D project selection (13 *criteria*).

Initially based on the critical review of the above literature, 6 categories of criteria and 28 specific sub-criteria are identified and considered in the prioritization of projects portfolio for the present research work, as determined by the expert team. These six criteria are as follows: commercial matters (C_1) with *six* sub-criteria, project selection risk (C_2) with *eight* sub-criteria's, project owners (C_3) with *four* sub-criteria, project bid competition (C_4) with *three* sub-criteria, additional factors (C_5) with *three* sub-criteria, internal operating issue (C_6) with *four* sub-criteria's. The detailed criteria and sub-criteria considered for project portfolio selection is shown in Table 2, along with detail description of sub-criteria in Table 3 respectively.

Table 2 Considered criteria in proposed project portfolio selection method

Criteria	Sub-criteria with references
C_1: Commercial matters	Project Budget [59]; Expected return on investment [56]; Payback period [58]; Potential market share [55]; Ethics [6]; Initial Cash Outlay [53]
C_2: Project selection risk	Fund Arrival Rate [3]; Technical Risk [57]; Ratio of Completion Time And Avg. [1]; Financial Risk [55]; Quality Risk [22]; Transportation availability [54]; Legal Exposure [22]; Political Impact In Area [55]
C_3: Project owners	Availability of workforce [55]; Management Attitude [52]; Work Experience Of Project Managers [54]; Owner's Policies And regulations [52]
C_4: Project bid competition	Number Of Competitors [55]; Monitoring and feedback [55]; Competitors Strategy [52]
C_5: Additional factors	Patent Protection [53]; Impact On Company Image [53]; Strategic Fit [55]
C_6: Internal operating issue	Need To Train Employee [52]; Change In Workforce Size [54]; Change In Physical Environment [53]; Change In service operations [53]

4.3 Detailed description of alternative project proposal

For selection of projects proposals, the experts consulted various literature papers. Tavana et al. [2] proposal a fuzzy hybrid project portfolio selection method using DEA, TOPSIS for evaluation of projects grouped under clean fuel activity, producing *petrochemical products* and hydrogen. Costantino et al. [61] developed an artificial neural network (ANN) model based on critical success factors for *electrical transmission* based company. Altuntas and Dereli [62] proposed a DEMATEL cum patent citation analysis method for prioritizing a portfolio of investments projects in *telecommunication sector*. Bhattacharya [63] projected a case study of an organization involved in different structural works in civil, mechanical and *electrical field* using grey theory based MCDM approach. Huang et al. [22] proposed a fuzzy AHP model in R&D project selection in *telecommunication, chemical* engineering and pharmaceuticals.

Based on the above paper and in Sect. 4.2, the experts after common consensus among them, selected four project proposals viz. *construction of building infrastructure sectors* (PP1), *transport infrastructure sectors* (PP2), *power transmission and distribution sectors* (PP3), *hydrocarbon and chemical sectors* (PP4) and *telecommunication sectors* (PP5). The detail descriptions are given in Tables 4 and 5.

5 Computational procedure using the proposed approach

A hypothetical example of project evaluation by a panel of decision makers (Table 1) using the fuzzy AHP based project selection model is presented in Fig. 1. In the *first or top level* is the overall goal of the system—*To obtain portfolio of project on priority bases*. The *second level* is made up of six criteria, which contribute to the goal and *third level* is made up of the sub-criteria constituting each

Table 3 Detailed description of sub-criteria for project portfolio selection

Sub-Criteria	Description
C_{11}: Project budget	The total estimated cost (*recurring and non-recurring*) associated with the Project. It is an extremely important criterion for systematic planning and problem formulation
C_{12}: Expected return on investment	Tool to determine whether or not an investment has a positive or negative average net outcome. It is not a hard and fast figure of profit and loss
C_{13}: Pay back period	The intent of project payback period is to estimate the amount of time that will be necessary to re-coup the investment in a project, i.e. the time taken for the project to payback its initial budget and generate positive cash flow for the company
C_{14}: Potential market share	Portfolio balance supports developing a strategy that allows companies to balance or offset risk; explore alternative market opportunities; and/or allows the funding of innovation in other product lines
C_{15}: Ethics	Ethical issues can limit choices for research method. Ethics risk is low but ethics clearance is required due to involvement of participants. In long run, this criterion is crucial for company's image in the market
C_{16}: Initial cash outlay	Before prioritization of projects proposals, the company should see the initial cash outlay so that the project can be completed in due time. Discounted cash flow (DCF) method estimate cash outlays and expected cash inflow resulting from investment
C_{21}: Fund arrival rate	Project managers should make paper works ready (Utilization certificate) for fresh allotment of funds
C_{22}: Technical risk	Risk due to the development of new or untested technologies. Availability of the required technology and expertise to accomplish the specific technical action steps
C_{23}: Ratio of completion time and avg.	The project manager should have sound knowledge of the expected completion rate with the average time taken, for financial and human resource constraints
C_{24}: Financial risk	Cost over budget, risks from the financial exposure caused by investing in the Project. Financial criteria are subject to assumptions due to level of uncertainty and time of project selection
C_{25}: Quality risk	Risks due to firm's goodwill or reputation due to the quality of the completed project. For investment projects
C_{26}: Transportation availability	It shows the traffic comfort where the project is implemented
C_{27}: Legal exposure	Potential for law suits or logical obligation, instructions must be clear and comprehensive so if any problem is occurred, it must be foreseen
C_{28}: Political impact in the area	Some Projects or some areas have political pressures that cause regional unrest during implementation of Project
C_{31}: Availability of workforce	The consideration of this critical resource is proposed in conjunction with risk and project relevance. There are key people in an organization who have involvement in most projects, and their availability will represent critical path for those projects. So, their availability will be a major determinant to the projects that can be selected both from economic and competence views
C_{32}: Management attitude	The management's behavior and dealing with workers can be motivated to select the projects. It includes political acceptance, end-user understanding, co-operation and commitment to the project and matching with users' interest or workload. Logical behavior causes owner to contribute more about projects

Table 3 continued

Sub-Criteria	Description
C_{33}: Experience of project managers	Project manager's experience in similar projects can remove problems for the owner and provide better recommendation for implementing the project. The extent to which project managers are ready to cope up and work under the change in worksite environment for public safety
C_{34}: Owner's policies and regulations	The owner's policies may cause to prevent some harm of projects. This is one of a positive criterion that can increase profit
C_{41}: Number of competitors	Size and number of competitors in market for project based on proposed technology
C_{42}: Monitoring and feedback	Timely provision of the comprehensive control information at each stage in the implementation process
C_{43}: Competitors strategy	Company should be well-aware of the strategic policy of the competitors and market strategy to form their own strategy preventing leakage in information from the company's interior, which fits best upper edge in the market
C_{51}: Patent protection	The strategies and new technologies formed by research team of the company should be patented to prevent leakage in information and staying apart in this global competitive market
C_{52}: Impact on company image	Companies should have clearly defined goals and better financial dealing with previous projects creating good reputation in market for getting future projects
C_{53}: Strategic fit	Strategic fit aimed at evaluating the impact of the project on the position of firm in the competitive context. It is aligned with requirement analysis which helps define the initial project choices
C_{61}: Need to train employee	Recruitment, selection and training of the necessary personnel for the project team, for change in project technologies. This enhances the awareness of potential risks
C_{62}: Change in work force size	Work force should not be stagnant in any project site for longer duration, as they can be aware of the policies and strategies of the company, which can be secretly passed to competitors companies
C_{63}: Change in physical environment	This criterion contains a set of geography and climate parameters in project location that should be implemented. It helps to handle unexpected crises and deviations from plan
C_{64}: Change in service operations	The extent to which proposed service is potential for further technology development based on project results

criteria for project proposal evaluation. The details given in Sect. 4.2. *Fourth level* is the evaluation of candidate projects (*discussed in* Sect. 4.3).

5.1 Determination of weights of criteria and sub-criteria applying fuzzy AHP

The procedure starts with the elicitation of pair-wise comparison judgments to set priorities for the criteria used in the evaluation. The second level elements (criteria) in Fig. 1, are arranged into matrix and the pair-wise comparisons are done using the fuzzy AHP judgment scale (Table 7). The question for each entry in the comparison matrix is: 'When the element on the left hand side is compared with the element on

Table 4 List of alternatives project selection in portfolio management

Alternatives	Project proposal	Project areas (*Sectors*)
PP1	Project proposal 1	Housing infrastructure sectors
PP2	Project proposal 2	Transport Infrastructure sectors
PP3	Project proposal 3	Power transmission and distribution sectors
PP4	Project proposal 4	Hydrocarbon and chemical sectors
PP5	Project proposal 5	Telecommunication sectors

the top, how much more (or less) important it is with respect to the overall goal of selecting the best portfolio of projects?' For the answer, the following rules (Rules 1–3) are applicable:

Rule 1 If the elements on the left hand side are more important, it is given linguistic importance value (Table 6), then converting to TFNs corresponding to its importance.

Rule 2 If the elements are less important, it is given corresponding reciprocal value of the corresponding TFNs.

Rule 3 Of all the pairs in the matrix, only the pairs written above the diagonal need to be compared, as the elements below the diagonal of the matrix are obtained by taking the reciprocal of the upper ones.

Thus the pair-wise comparison of the six criteria in relation in respect of goal is done to reflect the importance of each of the criteria (Table 7). As explained, the fuzzy AHP determines the priorities of each criteria or (*the importance of weights that should be given to each criteria*) analyzing such judgmental matrices, using *Chang's extent analysis* [38] as detailed in Sect. 3.2.

Next, the pair-wise comparison matrices are developed to reflect the importance of *each sub-criterion* in relation to the *six criteria* on an individual basis (see Tables 8, 9, 10, 11, 12, 13).

After combining the priorities over all hierarchy, overall importance weights of criteria (Table 14) and sub-criteria (second *column of* Table 15) are determined.

Global hierarchical priority weight for the sub-criteria are obtained by multiplying local weight of sub-criteria (each of the *level* 3 priority) by the corresponding local weight of criteria (*level* 2 priority), as shown in *third column of* Table 15. The corresponding ranking of sub-criteria is given in *column four* of Table 15.

5.2 Evaluation and prioritization of projects proposal in project portfolio

Next step is to establish the composite priorities for the candidate projects, if the number of project is small. However, for large number of project alternatives, it is generally infeasible. For small number of project proposals, pair wise comparison among them is done, compared with respect to each criterion. The explosion in the number of required comparisons is the basic criticism of the AHP method. To overcome this problem, *absolute measurement method* in AHP proposed by Kumar [60] is done where the alternative projects are not compared pair-wise but rated

Table 5 Detailed description of sub-criteria for project portfolio selection

Project areas (*Sectors*)	Description of the project areas
Housing infrastructure sectors (PP1)	To provide affordable housing is a major challenge in regard to increasing population. This project aim to initiate development of new housing scheme keeping in view sustainability, green carbon emission and green building category. These projects are based on PPP models (http://www.mwhglobal.com). This project lifecycle starts from initial project investigation through to maintenance and operational management network
Transport infrastructure sectors (PP2)	Engineering consulting for transport infrastructure projects is an important part of service project portfolio. Areas of activity include metros, suburban & high speed railways, airports and harbors. These projects are based on public–private-partnership (PPP) model which involves the development of infrastructure projects by private sector players (*Companies*) in partnerships with existing government (http://www.lntidpl.com).
Power transmission and distribution sectors (PP3)	Growth of the Power sector depends on power generation capacity, upgradation of existing lines & inter regional connectivity, thus a reliable evacuation and transmission system to keep us ahead of the times. A proper design for its production, channelization and distribution will meet the growing demands and propel our economy forward (http://esselinfraprojects.com/vcpowertransmision.php)
	Distribution projects including railway electrification and project electrification in the domestic and international markets
Hydrocarbon and chemical sectors (PP4)	The projects in refining and petrochemical sectors are generally massive projects with high level of multi-disciplinary complexities involved. In the overseas markets, many customers have found it beneficial to transfer entire project risk and responsibility to competent EPC contractors, thereby eliminating the need for a large management team for day-to-day monitoring, integration and management of engineering, procurement and construction functions (http://www.essar.com). The hectic activities in the hydrocarbon sector coupled with government's pro-active approach for the sector provides ample scope for engineering, procurement and construction (EPC) companies to grow
Telecommunication sectors (PP5)	These project sector is characterized by high rates of innovation associated with a vast array of sustainability risks and challenges. Shorter product life cycles and escalated customers' expectation adds complexity to the business environment. Globally, the development of telecommunication projects is rapidly increasing with one innovation replacing in a matter of years or even weeks (http://blogs.dnvgl.com/sustainability/2012). Many companies have placed great emphasis to enhance the integration and collaborate effort with different supplier counterparts to increase visibility in this sector

against each criterion on the suitable grading scale (Table 17) with five elements. Few works has been done in this area [64, 65].

For *fourth level* in the hierarchy (Fig. 1), list of alternative project proposals in project portfolio, after discussion with decision makers, detailed in Tables 4 and 5. Experts gave a grading scale with five elements for the prioritization of particular

Table 6 Triangular fuzzy conversion scale

Linguistic importance value	Fuzzy pair wise comparison value
Just equal (JE)	(1, 1, 1)
Equally important (EI)	(1, 1, 3)
Weakly important (WI)	(1, 3, 5)
Moderately important (MI)	(3, 5, 7)
Strongly important (SI)	(5, 7, 9)

Table 7 The Fuzzy evaluation of criteria in respect to desired goal

Goal	C_1	C_2	C_3	C_4	C_5	C_6	W_0
C_1	(1,1,1)	(1,1,3)	(1,3,5)	(3,5,7)	(1,3,5)	(1,1,3)	0.22
C_2		(1,1,1)	(1,3,5)	(1,1,3)	(5,7,9)	(1,3,5)	0.24
C_3			(1,1,1)	(5,7,9)	(1,1,3)		0.18
C_4				(1,1,1)		(3,5,7)	0.12
C_5				(3,5,7)	(1,1,1)	(1,1,3)	0.16
C_6			(1,1,3)			(1,1,1)	0.08

The weight vector from Table 1 is calculated as $W_0 = (0.22, 0.24, 0.18, 0.12, 0.16, 0.08)^T$

Table 8 The Fuzzy evaluation of sub-criteria in respect to criteria C_1

	C_{11}	C_{12}	C_{13}	C_{14}	C_{15}	C_{16}	W_0
C_{11}	(1,1,1)		(1,1,3)	(1,1,3)			0.14
C_{12}	(1,1,3)	(1,1,1)	(1,1,3)			(1,3,5)	0.17
C_{13}			(1,1,1)			(1,1,3)	0.12
C_{14}		(1,3,5)	(1,3,5)	(1,1,1)	(1,1,3)	(3,5,7)	0.25
C_{15}	(1,3,5)	(1,1,3)	(1,3,5)		(1,1,1)	(3,5,7)	0.23
C_{16}	(1,1,3)					(1,1,1)	0.09

Table 9 The Fuzzy evaluation of sub-criteria in respect to criteria C_2

	C_{21}	C_{22}	C_{23}	C_{24}	C_{25}	C_{26}	C_{27}	C_{28}	W_0
C_{21}	(1,1,1)	(1,1,3)	(1,3,5)	(3,5,7)	(1,3,5)	(1,1,3)	(1,3,5)	(1,1,3)	0.18
C_{22}		(1,1,1)	(1,3,5)	(1,1,3)	(5,7,9)	(1,3,5)	(1,1,3)	(1,3,5)	0.20
C_{23}			(1,1,1)	(5,7,9)	(1,1,3)		(1,1,3)	(3,5,7)	0.17
C_{24}				(1,1,1)		(3,5,7)	(1,3,5)	(1,3,5)	0.15
C_{25}				(3,5,7)	(1,1,1)	(1,1,3)	(1,3,5)	(1,1,1)	0.14
C_{26}			(1,1,3)			(1,1,1)		(1,1,3)	0.08
C_{27}						(1,1,3)	(1,1,1)	(1,1,1)	0.03
C_{28}					(1,1,1)		(1,1,1)	(1,1,1)	0.05

project portfolio, as shown in Table 17. Normally, grades are assumed having same value for each of the parent criteria. In this paper, on the basis of expert's opinion on each project alternative and each criterion, we prepare a judgment matrix of the grades priority (Table 16).

Table 10 The Fuzzy evaluation of sub-criteria in respect to criteria C_3

	C_{31}	C_{32}	C_{33}	C_{34}	W_0
C_{31}	(1,1,1)	(1,1,3)	(1,1,3)	(1,3,5)	0.30
C_{32}		(1,1,1)	(1,3,5)	(1,3,5)	0.33
C_{33}			(1,1,1)		0.17
C_{34}			(1,1,3)	(1,1,1)	0.20

Table 11 The Fuzzy evaluation of sub-criteria in respect to criteria C_4

	C_{41}	C_{42}	C_{43}	W_0
C_{41}	(1,1,1)			0.22
C_{42}	(1,3,5)	(1,1,1)	(1,1,3)	0.45
C_{43}	(1,1,1)		(1,1,1)	0.33

Table 12 The Fuzzy evaluation of sub-criteria in respect to criteria C_5

	C_{51}	C_{52}	C_{53}	W_0
C_{51}	(1,1,1)	(1,3,5)		0.37
C_{52}		(1,1,1)		0.06
C_{53}	(1,3,5)	(3,5,7)	(1,1,1)	0.57

Table 13 The Fuzzy evaluation of sub-criteria in respect to criteria C_6

	C_{61}	C_{62}	C_{63}	C_{64}	W_0
C_{61}	(1,1,1)		(1,1,1)	(1,1,3)	0.26
C_{62}	(1,1,3)	(1,1,1)	(1,3,5)	(1,1,3)	0.32
C_{63}			(1,1,1)		0.18
C_{64}			(1,1,3)	(1,1,1)	0.24

Table 14 The ranking of fuzzy weight of criteria

Criteria	Weightage	Ranking
Commercial matters (C_1)	0.22	2
Project selection risk (C_2)	0.24	1
Project owners (C_3)	0.18	4
Project bid competition (C_4)	0.12	3
Additional factors (C_5)	0.16	5
Internal operating issue (C_6)	0.08	6

Candidate projects are evaluated for each criterion by identifying the grade (Table 17) which best describes it and the priority score obtained for each project proposals in portfolio are obtained and ranked based on their magnitude (Table 18).

Table 15 The weights of supplier selection criteria in investment project hierarchy

Criteria with local wt.	Sub-criteria with local wt.	Global priority wt.	Ranking
C_1: Commercial matters (0.22)	C_{11}: Project budget (0.14)	0.0308	16
	C_{12}: Expected investment return (0.17)	0.0374	13
	C_{13}: Payback period (0.12)	0.0264	18
	C_{14}: Potential market share (0.25)	0.0550	4
	C_{15}: Ethics (0.23)	0.0506	7
	C_{16}: Initial cash outlay (0.09)	0.0198	23
C_2: Project selection risk (0.24)	C_{21}: Fund arrival rate (0.18)	0.0432	9
	C_{22}: Technical risk (0.20)	0.0480	8
	C_{23}: Ratio of completion time and avg. (0.17)	0.0408	10
	C_{24}: Financial risk (0.15)	0.0360	14
	C_{25}: Quality risk (0.14)	0.0336	15
	C_{26}: Transportation availability (0.09)	0.0216	21
	C_{27}: Legal exposure (0.07)	0.0168	25
	C_{28}: Political impact in the area (0.05)	0.0120	27
C_3: Project owners (0.18)	C_{31}: Availability of workforce (0.30)	0.0540	5
	C_{32}: Management attitude (0.32)	0.0576	3
	C_{33}: Work experience of project manager (0.17)	0.0306	17
	C_{34}: Owners policies and regulations (0.21)	0.0378	12
C_4: Project bid competition (0.12)	C_{41}: Number of competitors (0.22)	0.0261	19
	C_{42}: Monitoring and feedback (0.45)	0.0535	6
	C_{43}: Competitors strategy (0.33)	0.0392	11
C_5: Additional factors (0.16)	C_{51}: Patent Protection (0.37)	0.0592	2
	C_{52}: Impact on company image (0.06)	0.0096	28
	C_{53}: Strategic fit (0.57)	0.0912	1
C_6: Internal operating Issue (0.08)	C_{61}: Need to train employee (0.26)	0.0208	22
	C_{62}: Change in work force size (0.32)	0.0250	20
	C_{63}: Change in physical environment (0.18)	0.0144	26
	C_{64}: Change in service operations (0.24)	0.0192	24

Subjective judgments on expected outcomes and intended impacts are typical in project portfolio selection by the management. As per unanimous choice from decision makers' side, the result of project proposals in a portfolio in respect to different sub-criteria is evaluated, shown in Table 18.

For example, for *project proposal* 1 (PP1) the experts (*based on majority decision*) concludes that the project budget is at "G" (Good) grade, *expected investment return* at "E" (Excellent) grade etc. Similarly, the grading values of other four *project proposals* PP2, PP3, PP4 and PP5 are obtained, and detailed in Table 18. We then calculate the scores for each project proposals. The final ranking

Table 16 Fuzzy AHP Comparison Matrix of linguistic grading (*for level* 4 *Alternative*)

	V	P	S	G	E	Priority
V	(1,1,1)	(1/5, 1/3,1)	(1/7,1/5,1/3)	(1/9,1/7,1/5)	(1/9,1/9,1/7)	0.015
P	(1,3,5)	(1,1,1)	(1/5, 1/3,1)	(1/7,1/5,1/3)	(1/9,1/7,1/5)	0.111
S	(3,5,7)	(1,3,5)	(1,1,1)	(1/5,1/3,1)	(1/7,1/5,1/3)	0.162
G	(5,7,9)	(3,5,7)	(1,3,5)	(1,1,1)	(1/5,1/3,1)	0.301
E	(7,9,9)	(5,7,9)	(3,5,7)	(1,3,5)	(1,1,1)	0.411

V very poor, *P* poor, *S* satisfactory, *G* good, *E* excellent

Table 17 Linguistic grading scale with priority values for selecting project alternatives

Linguistic grading scale	Priorities
Very poor (V)	0.015
Poor (P)	0.111
Satisfactory (S)	0.162
Good (G)	0.301
Excellent (E)	0.411

is shown in Table 19, based on which the suggested methodology specify that PP1 is the first priority in project portfolio with total score of 0.2827.

In the application it is shown that calculation of criteria and sub-criteria weights are important in fuzzy AHP method and could alter the ranking for other projects.

6 Result discussion

As per Table 14, the ranking of fuzzy weight of criteria are given. Project selection risk (C_2) is given *maximum weight* (0.24) followed by commercial matters (C_1) with score point 0.22. Additional factors (C_5) and internal operating Issue (C_6) are given low score with 0.16 and 0.08, respectively. Table 15 gives the local weight of sub-criteria along with its global priority weight (3rd *column of* Table 15) in portfolio prioritization hierarchy. Twenty-eight sub-criteria classified into six criteria are ranked accordingly (4th *column of* Table 15). The local weight of the sub-criteria is multiplied with respective criteria weight to get the global priority weight of the sub-criteria. Here we take the criteria along with the sub criteria and discuss in details about their ranking.

- For *first* criteria *Commercial Matter* (C_1), Sub-criteria like *Ethics* (C_{15}) and *Potential market share* (C_{14}) have greater weight age among six other sub-criteria with score point of 0.0506 and 0.0550 respectively, getting 7th and 4th rank in global priority ranking.
- For *second* criteria *Project selection risk* (C_2), sub-criteria like *Technical risk* (C_{22}) and *Fund arrival rate* (C_{21}) dominated the other eight sub-criteria's with score point of 0.0480 and 0.432 respectively, getting 8th and 9th ranking in global priority.

Table 18 Experts Grade on different Project Proposals

Sl. no.	Sub-criteria	Wt.	PP_1	PP_2	PP_3	PP_4	PP_5
C_{11}	Project budget (0.14)	0.0308	S	E	S	G	G
C_{12}	Expected investment return (0.17)	0.0374	E	S	P	S	G
C_{13}	Payback period (0.12)	0.0264	P	G	G	G	P
C_{14}	Potential market share (0.25)	0.0550	G	P	P	P	P
C_{15}	Ethics (0.23)	0.0506	E	S	S	V	G
C_{16}	Initial cash outlay (0.09)	0.0198	S	E	E	G	V
C_{21}	Fund arrival rate (0.18)	0.0433	P	V	P	V	P
C_{22}	Technical risk (0.20)	0.0481	G	G	P	P	G
C_{23}	Ratio of completion time and avg. (0.17)	0.0408	V	P	G	G	G
C_{24}	Financial risk (0.15)	0.0360	P	G	G	S	P
C_{25}	Quality risk (0.14)	0.0336	G	P	P	S	S
C_{26}	Transportation availability (0.09)	0.0216	S	G	E	G	S
C_{27}	Legal exposure (0.07)	0.0168	P	S	S	P	V
C_{28}	Political impact in the area (0.05)	0.0120	G	P	E	V	P
C_{31}	Availability of workforce (0.30)	0.0540	E	E	G	P	S
C_{32}	Management attitude (0.32)	0.0576	S	P	P	S	G
C_{33}	Work experience of project manager (0.17)	0.0306	E	S	P	S	G
C_{34}	Owners policies and regulations (0.21)	0.0378	E	S	P	S	G
C_{41}	Number of competitors (0.22)	0.0261	G	P	S	P	V
C_{42}	Monitoring and feedback (0.45)	0.0535	E	S	G	P	S
C_{43}	Competitors strategy (0.33)	0.0392	E	G	S	G	E
C_{51}	Patent protection (0.37)	0.0592	E	G	S	G	S
C_{52}	Impact on company image (0.06)	0.0096	V	S	G	G	S
C_{53}	Strategic fit (0.57)	0.0912	G	G	G	G	G
C_{61}	Need to train employee (0.26)	0.0208	G	G	V	S	E
C_{62}	Change in work force size (0.32)	0.0256	G	G	P	E	S
C_{63}	Change in physical environment (0.18)	0.0144	S	P	V	G	G
C_{64}	Change in service operations (0.24)	0.0192	P	S	S	G	S

Table 19 Total scores of project proposal

Project proposals	Total score	Preference of project selection
PP1	0.2827	1
PP2	0.2223	2
PP3	0.1963	4
PP4	0.1933	5
PP5	0.2217	3

- For *third* criteria, *Project owners* (C_3), sub-criteria's like *Management Attitude* (C_{32}) and *Availability of workforce* (C_{31}) gained score point of 0.0576 and 0.0540 respectively, with global ranking of 3rd and 5th place.

- For *fourth* criteria, *Project bid competition* (C_4), sub-criteria like Monitoring and feedback (C_{42}) and Competitors strategy (C_{43}) in spite of earning score point of 0.0535 and 0.0392 respectively ranked 6th and 11th places.
- For *fifth* criteria, namely, *Additional factors* (C_5), sub-criteria like *strategic fit* (C_{53}) and *patent protection* (C_{51}) has a healthy score point of 0.0912 and 0.0592, topping the chart with 1st and 2nd in global scenario.
- For *sixth* criteria, namely, *Internal Operating Issue* (C_6), sub-criteria like *Change in workforce size* (C_{62}) and *Need to train employee* (C_{61}) ranked 20th and 24th with score point of 0.0250 and 0.0208, respectively.

Based on the fuzzy comparison matrix (Table 16) of '*grading for experts*' the priority weights followed by the expert grading on different project proposals, along with the global weights of the sub-criteria, total score points of the five project proposals are sorted out, shown in Table 18 and ranking given in Table 19.

7 Concluding remarks

Prioritization of appropriate set of projects portfolio with improved consequence, risk and cost is perilous problem which decision makers are currently facing. In this paper, Chang's extent analysis based fuzzy AHP is practically applied in prioritization of project portfolio problem. The project portfolio prioritization is a procedure for assessing the appropriate investment and choose the best project proposal through multi-criteria mode uses linguistic preferences for uncertain situations. Fuzzy AHP examine the hierarchy of the project selection problem, decide the weights of the criteria, and achieve the project preference ranking in portfolio management.

Considering a set of supportive criteria and sub-criteria through an exhaustive literature review, a broad project selection framework is proposed for project selection problem. In typical MCDM problematic, numerous DMs views has direct effect on calculation on importance of criteria. The conflictive DMs preferences on criteria gathered through linguistic terms, is parameterized using TFNs. In MCDM field, there are several crisp or fuzzy ranking methods that only provide DMs with a ranking order of alternatives. In the method applied in this paper, the decision makers use a clear list of the linguistic terms and weights attributed to them thus providing both ranking and weightage information to the investors. Fuzzy numbers mathematics is the bearing configuration of the language and of the procedure.

In this portfolio assortment, final fuzzy scores are assessed for different α-cuts. It is obvious that, when the level of α-cut is equal to 1, the fuzzy AHP converges to crisp AHP. Thus, working with fuzzy numbers and its operations offer great flexibility in decision-making. With the help of pre-determined criteria, this decision methodology assist experts to prioritize among the project portfolios by eliminating inefficient one's in indeterminate resource, outcome and success probability.

In future studies this method can be executed on big data set in fuzzy stochastic situations taking fuzzy random parameters and interrelationships among criteria. Nature inspired computing like, NEGA (*Nondominated sorting genetic algorithm*),

NPGA (*niched Pareto genetic algorithm*), PAES (*Pareto archived evolution strategy*) and PSO (*particle swarm optimization*) can be engaged to solve the problem, for large data set.

References

1. Meredith, J., Mantel, S., Shafer, S.: Project Management: A Managerial Approach, 9th edn (2015)
2. Tavana, M., Keramatpour, M., Santos-Arteaga, F., Ghorbaniane, E.: A fuzzy hybrid project portfolio selection method using Data Envelopment Analysis, TOPSIS and Integer Programming. Expert Syst. Appl. **42**, 8432–8444 (2015)
3. Turner, J.: The Handbook of Project-Based Management, 3rd edn (2009)
4. Wang, J., Xu, Y., Li, Z.: Research on project selection system of pre-evaluation of engineering design project bidding. Int. J. Proj. Manag. **27**, 584–599 (2009)
5. Kaiser, M., El Arbi, F., Ahlemann, F.: Successful project portfolio management beyond project selection techniques: understanding the role of structural alignment. Int. J. Proj. Manag. **33**, 126–139 (2015)
6. Archer, N., Ghasemzadeh, F.: An integrated framework for project portfolio selection. Int. J. Proj. Manag. **17**(4), 207–216 (1999)
7. Saaty, T.: The analytic hierarchy process. McGraw-Hill, New York (1980)
8. Shouder, W.: A scoring methodology for assessing the suitability of management science models. Manag. Sci. **18**(10), 526–543 (1972)
9. Amiri, M.: Project selection for oil-fields development by using the AHP and fuzzy TOPSIS methods. Expert Syst. Appl. **37**, 6218–6224 (2010)
10. Ghapanchi, A., Tavana, M., Khakbaz, M., Low, G.: A methodology for selecting portfolios of projects with interactions and under uncertainty. Int. J. Proj. Manag. **30**, 791–803 (2012)
11. Costantino, F., Gravio, G., Nonino, F.: Project selection in project portfolio management: an artificial neural network model based on critical success factors. Int. J. Proj. Manag. **33**(8), 1744–1754 (2015)
12. Mohanty, R., Agarwal, R., Choudhury, A., Tiwari, M.: A fuzzy ANP-based approach to R&D project selection: a case study. Int. J. Prod. Res. **43**(24), 5199–5216 (2005)
13. Chu, P., Hsu, Y., Fehling, M.: A decision support system for project portfolio selection. Comput. Ind. **32**(2), 141–149 (1996)
14. Gutjahr, W., Katzensteiner, S., Reiter, P., Stummer, C., Denk, M.: Competence-driven project portfolio selection, scheduling and staff assignment. CEJOR **16**, 281–306 (2008)
15. Shpak, A., Zaporojan, D.: Working out R&D program via multi-criteria analysis. Comput. Sci. Moldova **4**, 239–259 (1996)
16. Danila, N.: Strategic evaluation and solution of R&D projects. R&D Manag. **19**, 47–62 (1999)
17. Halouani, N., Chabchoub, H., Martel, J.: Promethee-MD-2T method for project selection. Eur. J. Oper. Res. **195**, 841–849 (2009)
18. Tiryaki, F., Ahlatcioglu, B.: Fuzzy portfolio selection using fuzzy analytic hierarchy process. Inf. Sci. **179**(1), 53–69 (2009)
19. Wang, J., Hwang, W.: A fuzzy set approach for R&D portfolio selection using a real options valuation model. Omega **35**, 247–257 (2007)
20. Coffin, M., Taylor, B.: Multiple criteria R&D project selection and scheduling using fuzzy logic. Comput. Oper. Res. **23**(3), 207–220 (1996)
21. Chen, C., Cheng, H.: A comprehensive model for selecting information system project under fuzzy environment. Int. J. Proj. Manag. **27**, 389–399 (2009)
22. Huang, C., Chu, Y., Chiang, Y.: A fuzzy AHP application in government-sponsored R&D project selection. Omega **36**(6), 1038–1052 (2008)
23. Patanakul, P., Shenhar, A., Milosevic, D.: How project strategy is used in project management: cases of new product development and software development projects. J. Eng. Technol. Manag. **29**, 391–414 (2012)
24. Lu, V., Kim, S., Huynh, T.: Improving project management performance of large contractors using benchmarking approach. Int. J. Proj. Manag. **26**, 758–769 (2008)
25. Ika, L.: Project success as a topic in project management journals. Proj. Manag. J. **40**, 6–19 (2009)

26. Project Management Institute. The standard for portfolio management, 3rd edn. 14 Campus Boulevard, Newtown Square, Pennsylvania 19073-3299 USA, Project Management Institute, Inc, 189 (2013)
27. Mohanty, R.: Project selection by a multiple-criteria decision-making method: an example from a developing country. Int. J. Proj. Manag. **10**, 31–38 (1992)
28. Turner, J., Müller, R.: The project manager's leadership style as a success factor on projects: a literature review. Proj. Manag. J. **36**, 49–61 (2005)
29. Salo, A., Keisler, J., Morton, A.: An invitation to portfolio decision analysis. In: Salo, A., Keisler, J., Morton, A. (eds.) Portfolio decision analysis: improved methods for resource allocation, pp. 3–27. Springer, New York (2011)
30. Carlsson, C., Fuller, R., Heikkila, M., Majlender, P.: A fuzzy approach to R&D project portfolio selection. Int. J. Approx. Reason. **44**, 93–105 (2007)
31. Ho, W., Tsai, C., Tzeng, G., Fang, S.: Combined DEMATEL technique with a novel MCDM model for exploring portfolio selection based on CAPM. Expert Syst. Appl. **38**, 16–25 (2011)
32. Jeng, D., Huang, K.: Strategic project portfolio selection for national research institutes. J. Bus. Res. **68**, 2305–2311 (2015)
33. Khalili-Damghani, K., Sadi-Nezhad, S.: A hybrid fuzzy multiple criteria group decision making approach for sustainable project selection. Appl. Soft Comput. **13**, 339–352 (2013)
34. Liesio, J., Punkka, A.: Baseline value specification and sensitivity analysis in multiattribute project portfolio selection. Eur. J. Oper. Res. **237**, 946–966 (2014)
35. Abbasi, M., Ashrafi, M., Tashnizi, E.: Selecting balanced portfolios of R&D projects with interdependencies: a cross-Entropy based methodology. Technovation **34**, 54–63 (2014)
36. An, D., Yang, Y., Xi, B., Dong, L., Ren, J.: Mitigating pollution of hazardous materials from WEEE of China: portfolio selection for a sustainable future based on multi-criteria decision making. Resour. Conserv. Recycl. **105**, 198–210 (2015)
37. Smith-Perera, A., Garcia-Melon, M., Poveda-Bautista, R., Pastor-Ferrando, J.: A project strategic index proposal for portfolio selection in electrical company based on the analytic network process. Renew. Sustain. Energy Rev. **14**, 1569–1579 (2010)
38. Chang, D.: Applications of the extent analysis method on fuzzy AHP. Eur. J. Oper. Res. **95**, 649–655 (1996)
39. Zadeh, L.: Fuzzy sets. Inf. Control **8**, 338–352 (1965)
40. Zadeh, L.: The concept of a Linguistic variable and its application to approximate reasoning-I. Inf. Sci. **8**, 199–249 (1975)
41. Lin, C., Chen, Y.: Bid/no-bid decision-making-a fuzzy linguistic approach. Int. J. Proj. Manag. **22**, 585–593 (2004)
42. Saaty, T.: Decision making with dependence and feedback. The analytic network process. The organization and prioritization of complexity, 2nd edn. RWS Publications, Pittsburgh (2001)
43. Csutora, R., Buckley, J.: Fuzzy hierarchical analysis: the Lambda-Max method. Fuzzy Sets Syst. **120**, 181–195 (2001)
44. Van Laarhoven, P., Pedrycs, W.: A fuzzy extension of Saaty's priority theory. Fuzzy Sets Syst. **11**, 229–241 (1983)
45. Buckley, J.: Fuzzy hierarchical analysis. Fuzzy Sets Syst. **17**, 233–247 (1985)
46. Cheng, C., Mon, D.: Evaluation weapon system by AHP based on fuzzy scales. Fuzzy Sets Syst. **63**, 1–10 (1994)
47. Lee, A.: Application of modified fuzzy AHP method to analyze bolting sequence of structural joints. UMI Dissertation Service, A Bell & Howell Company (1995)
48. Cheng, C.: Evaluation naval tactical missile system by fuzzy AHP based on the grade value of membership function. European Journal of Operation Research **96**, 343–350 (1996)
49. Mikhailov, L.: Deriving priorities from fuzzy pair wise comparison judgments. Fuzzy Sets Syst. **134**, 365–385 (2003)
50. Chang, D.: Extent Analysis and Synthetic Decision, Optimization Techniques and Applications, vol. 1, pp. 352–355. World Scientific, Singapore (1992)
51. Gabriel, S., Kumar, S., Ordonez, J., Nasserian, A.: A multiobjective optimization model for project selection with probabilistic considerations. Socio Econ. Plan. Sci. **40**(4), 297–313 (2006)
52. Pinto, J., Slevin, D.: Critical success factors in R&D projects. Res. Technol. Manag. **32**, 31–35 (1989)
53. Meredith, J., Mantel, S.: Project Management: A Managerial Approach, 7th edn. Wiley, New York (2009)
54. Jiang, J., Klein, G.: Project selection criteria by strategic orientation. Inf. Manag. **36**, 63–75 (1999)

55. Puthamont, S., Charoenngam, C.: Strategic project selection in public sector: construction projects of the Ministry of Defense in Thailand. Int. J. Proj. Manag. **25**, 178–188 (2006)
56. Farrell, M.: Financial engineering in project management. Proj. Manag. J. **33**(1), 27–36 (2002)
57. Jafarizadeh, B., Khorshid-Doust, R.: A method of project selection based on capital asset pricing theories in a framework of mean-semi deviation behavior. Int. J. Proj. Manag. **26**, 612–619 (2008)
58. Lawson, C., Longhurst, P., Ivey, P.: The application of a new research and development project selection model in SMEs. Technovation **26**, 242–250 (2006)
59. De Maio, A., Verganti, R., Corso, M.: A multi-project management framework for new product development. Eur. J. Oper. Res. **78**, 178–191 (1994)
60. Kumar, S.: AHP-based formal system for R&D project evaluation. J. Sci. Ind. Res. **63**, 888–896 (2004)
61. Costantino, F., Gravio, G., Nonino, F.: Project selection in project portfolio management: an artificial neural network model based on critical success factors. Int. J. Proj. Manag. **33**, 1744–1754 (2015)
62. Altuntas, S., Dereli, T.: A novel approach based on DEMATEL method and patent citation analysis for prioritizing a portfolio of investment projects. Expert Syst. Appl. **42**, 1003–1012 (2015)
63. Bhattacharya, R.: A grey theory based multiple attribute approach for R&D project portfolio selection. Fuzzy Inf. Eng. **7**, 211–225 (2015)
64. Ohnishi, S., Saito, T., Yamanoi, T., Imai, H.: A weight representation for absolute measurement *AHP using fuzzy set theory*. In: 5th International Symposium on Computational Intelligence and Intelligent Informatics (ISCIII) (2011). https://doi.org/10.1109/isciii.2011.6069744
65. Felice, F., Petrillo, A.: Proposal of a structured methodology for the measure of intangible criteria and for decision making. Int. J. Simul. Process Model. **9**(3), 157–166 (2014)